Roadmap. Srednicki §3. pp. 1–2 ((3.1)–(3.6)) guess a relativistic free-particle H; pp. 3–12 derive it from an invariant $\mathcal{L}$: KG, modes with $\widetilde{dk}$, $[a, a^\dagger]$, zero-point energy. p. 5 replaces the missing hw 5–6; after p. 10 read p. 12 (hw 12); p. 11 (hw 13) is a rougher pass. Next: LecNote4 (§5) builds $|k\rangle = a^\dagger|0\rangle$. Supplements: ClassicalFields, FieldTransformations, DimensionalAnalysis_hbar, QFTHistory, FermionSigns.

See ~~Chapter~~ 3.

Canonical Quantization of Scalar Fields

(Also known as Second Quantization) [comment on this v.s Path-Integral].

Back to free particles.

$$H = \int d^3x\, a^\dagger(\vec{x}) \left(-\frac{1}{2m}\vec{\nabla}^2\right) a(\vec{x}).$$

$$= \int d^3p\, \tilde{a}^\dagger(\vec{p}) \left(\frac{1}{2m}\vec{p}^{\,2}\right) \tilde{a}(\vec{p})$$ ← with this easy transform to a Lorentzian (Respect special relativity easily?)

here $$\tilde{a}(\vec{p}) = \int \frac{d^3x}{(2\pi)^{3/2}}\, e^{-i\vec{p}\cdot\vec{x}}\, a(\vec{x})$$

You should all have seen such "Dual" Representation in QM.

Here I dropped $\hbar$ and c (we're in Natural Units)

$$\hbar = c = 1.$$

And we again have

$$[\tilde{a}(\vec{p}), \tilde{a}(\vec{p}')]_\mp = 0$$

$$[\tilde{a}^\dagger(\vec{p}), \tilde{a}^\dagger(\vec{p}')]_\mp = 0$$

$$[\tilde{a}(\vec{p}), \tilde{a}^\dagger(\vec{p}')]_\mp = \delta^3(\vec{p}-\vec{p}')$$

$$[A, B]_\mp = AB \mp BA.$$

$-$ for Bosons

$+$ for Fermions.

$\hat{a}^+(\vec{p})$ is the creation operator of a particle with momentum $\vec{p}$. and. annihilated by $\hat{a}(\vec{p})$

H is a theory of free particles.

Vacuum $|0\rangle$ has

$$\hat{a}(\vec{p})|0\rangle = 0$$

The other eigenstates of H are all of the form

$$\hat{a}^+(\vec{p}_1)\hat{a}^+(\vec{p}_2)\cdots\hat{a}^+(\vec{p}_n)|0\rangle$$

corresponds to n-particles with energy

$$E = E_1(\vec{p}_1) + E_2(\vec{p}_2) + \cdots + E_n(\vec{p}_n)$$

with $E_i(\vec{p}_i) = \frac{1}{2m}\vec{p}_i^{\,2}$

Now we can generalized it to relativistic one.

$$E(\vec{p}) = \sqrt{\vec{p}^2 + m^2}$$

we again can guess

$$H = \int d^3p\, (\vec{p}^2 + m^2)^{1/2}\, \hat{a}^+(\vec{p})\hat{a}(\vec{p})$$

now this is a relativistic theory of free particles. (spin zero)

Is it the theory Lorentz Invariant? Not in an Obvious way.

Let's see from a different view (emphasizing Lorentz Invariance)

Begin with a classical "Scalar field" $\varphi(x)$.

Scalar field means it is invariant under coordinate transformation.

$$\boxed{\varphi(x)} = \boxed{\bar{\varphi}(\bar{x})}, \quad \bar{x}^\mu = \Lambda^\mu{}_\nu x^\nu + a^\mu.$$

For a same field, two people at different inertial frames find identical values of them.

to avoid confusion, let us remove the bar on $\bar{\varphi}$ (in general $\varphi(x) \to \varphi'(x) = \bar{\varphi}(\bar{x})$)

This is in contrast to other covariant fields, such as vector fields

Given the scalar nature $\bar{\varphi}$ is the same function as φ.

we already know that

$$(-\partial^2 + m^2)\varphi(x) = 0.$$

recall: $(-\bar{\partial}^2 + m^2)\bar{\varphi}(\bar{x}) = 0$

The equation of motion (EoM) can be derived from variation of an action. $S = \int dt\, L$ here L is the Lagrangian

Action is local and can be written as

$$L = \int d^3x \underbrace{\mathcal{L}}$$

Lagrangian Density.

$$\rightarrow \quad S = \int dt\, L = \underline{\int d^4x\, \mathcal{L}}.$$

$\downarrow$

Lorentz invariant $\quad d^4\bar{x} = |\det \Lambda|\, d^4x = d^4x$

For the action S to be Lorentz invariant, we need $\mathcal{L}$ to be Lorentz invariant:

$$\tilde{\mathcal{L}}(\tilde{x}) = \mathcal{L}(x)$$

Let's construct it:

any simple function of $\varphi(x)$ would be L. Scalar:

L. Scalar

we can take

$$\mathcal{L}(x) = -\frac{1}{2}\partial^\mu\varphi(x)\,\partial_\mu\varphi(x) - \frac{1}{2}m^2\varphi^2(x) + \Omega_0$$

$[2]+[2]$; $+2$; $+2$; $+4$.

⇒ Dimensional Analysis

Teach students quickly. (Here declare Action S is dimensionless, actually $\hbar$). $\varphi(x)$ [1]

Euler Lagrange Equation. (E. o. M.)

$$0 = \delta S = \int d^4x\,\delta\mathcal{L}$$

$$= \int d^4x \left[-\frac{1}{2}\partial^\mu\delta\varphi\,\partial_\mu\varphi - \frac{1}{2}m^2\cdot 2\delta\varphi\varphi\delta\varphi - \frac{1}{2}\partial^\mu\varphi\,\partial_\mu\delta\varphi\right]$$

(same as textbook)

~~$= \int d^4x\,[-\partial^\mu\partial_\mu\varphi - m^2\varphi]\delta\varphi$~~

$$= \int d^4x\,[\partial^\mu\partial_\mu\varphi - m^2\varphi]\delta\varphi \quad \#$$

$$- \text{surface terms}\{-\delta\varphi(x)\,\partial\varphi(x)\big|_{x^\mu\to\pm\infty}$$

Include, say we restrict ourselves to $\delta\varphi = 0$ at boundary

see discussions in relativistic notes

← sloppy way of writing

Alternative ④-⑥

An alternative (quick solution to $\varphi(x)$). Skipping Pg ④-⑥

Recall we need to solve

$$(\partial^2 - m^2)\varphi(x) = 0 \quad \Rightarrow \quad (\vec{\partial}^2 - \partial_0^2 - m^2)\varphi(x) = 0$$

~~$\varphi(x) = \int d^3k / f(k)\, e^{-ikx}$~~

the solution should be as simple as

$$\varphi(x) = \int \frac{d^3k}{f(k)}\, a(\vec{k})\, e^{-ikx} + b(\vec{k})\, e^{+ikx}$$

where $kx = \vec{k}\cdot\vec{x} - \omega t$, $\omega = \pm\sqrt{\vec{k}^2 + m^2}$ ①

$$\varphi^*(x) = \int \frac{d^3k}{f(k)}\, a^*(\vec{k})\, e^{+ikx} + b^*(\vec{k})\, e^{-ikx}$$

So $\varphi(x) = \varphi^*(x)$ by reality condition

we find

$$a^*(\vec{k}) = b(\vec{k}), \quad b^*(\vec{k}) = a(\vec{k})$$

Hence

$$\varphi(x) = \int \frac{d^3k}{f(k)}\, a(\vec{k})\, e^{-ikx} + a^*(\vec{k})\, e^{+ikx}$$

Here we didn't spend much time discussing the energy term. We ~~might~~ encounter it again later.

1 Either root gives the same solutions: $\omega \to -\omega$, $\mathbf{k} \to -\mathbf{k}$ turns $a\,e^{ikx}$ into an e^{-ikx} term, so keeping both would count each mode twice. Take $\omega = +\sqrt{\mathbf{k}^2 + m^2}$ (p. 6's $\theta(k^0)$). Then e^{-ikx} is the negative frequency; its coefficient a^* becomes $a^\dagger$ (p. 10), which creates a particle of energy $+\omega$.

$kx = \vec{k}\cdot\vec{x} - \omega t = k^\mu x_\mu = k_\mu x^\mu = g^{\mu\nu} k_\mu x_\nu = g_{\mu\nu} k^\mu x^\nu$

$k^2 = k^\mu k_\mu = \vec{k}^2 - \omega^2 = -m^2.$

A four momentum k^μ that obeys $k^2 = -m^2$

is "on-shell" (on the mass ~~sell~~ shell).
in k^μ space
configuration)

Now let's see what makes

$$\int \frac{d^3k}{f(k)} \quad \text{L.I.} \quad \text{[and select the positive energy state]}$$

$S\alpha(1,3)\dagger$

Obviously d^4k is L.I.

$d^4k\, \delta(k^2+m^2)\, \theta(k^0)$ is L.I. for on-shell positive energy

L.I. L.I. Selecting the positive energy "orthochronous"

↑ Dirac Delta function.

↑ Unit step function

$$\int_{-\infty}^{\infty} dk^0\, \delta(k^2+m^2)\,\theta(k^0) = \frac{1}{2\omega}.$$

as $\int_{-\infty}^{\infty} dx\, \delta(g(x)) = \sum_i \frac{1}{|g'(x_i)|}$

$g(x)$ being any smooth function with simple zeros at $x = x_i$

thus for $f(k) \propto \omega$,

$d^3k/f(k)$ would be Lorentz Invariant (for on-shell states)

$$\widetilde{dk} \equiv \frac{d^3k}{(2\pi)^3 2\omega}.$$

Hence

$$\varphi(x) = \int \widetilde{dk} \left[a(\vec{k}) e^{ikx} + a^*(\vec{k}) e^{-ikx} \right].$$

We can get $a(\vec{k})$ via ~~Fourier~~ A general transformation (motivated by Fourier transformation between $\vec{x} \leftrightarrow \vec{p}$)

$$\int d^3x\, e^{-ikx} \varphi(x) = \frac{1}{2\omega} a(\vec{k}) + \frac{1}{2\omega} e^{2i\omega t} a^*(-\vec{k})$$

$$\int d^3x\, e^{-ikx} \partial_0 \varphi(x) = -\frac{i}{2} a(\vec{k}) + \frac{i}{2} e^{2i\omega t} a^*(-\vec{k})$$

Note: (3d and 4d)

$$\int_{-\infty}^{\infty} e^{ik(x-x')} \widetilde{dk} = (2\pi)^{(d)} \delta^{(d)}(x-x')$$

← for consistent notation.

$$\int_{-\infty}^{\infty} e^{i(k-k')x} d^dx = (2\pi)^{(d)} \delta^{(d)}(k-k')$$

$$\Rightarrow a(\vec{k}) = \int d^3x\, e^{-ikx} \left[i\partial_0 \varphi(x) + \omega \varphi(x) \right]$$

$$= i \int d^3x\, e^{-ikx} \overleftrightarrow{\partial_0} \varphi(x)$$

here $f \overleftrightarrow{\partial_\mu} g = f(\partial_\mu g) - (\partial_\mu f) g$, $\partial_0 \varphi = \frac{\partial \varphi}{\partial t} = \dot{\varphi}$

$a(\vec{k})$ is time-independent [1]

With Lagrangian, we can construct Hamiltonian.

Classical mechanics:

given $L(q_i, \dot{q}_i)$ of coordinate q_i and their time derivative, the conjugate momenta are given $p_i = \frac{\partial L}{\partial \dot{q}_i}$, and the Hamiltonian will be

$$H = \sum_i p_i \dot{q}_i - L.$$

Now q_i are replaced by $\varphi(x)$, x is an index (continuous)

the corresponding generalization would be

$$\Pi(x) = \frac{\partial \mathcal{L}}{\partial \dot{\varphi}(x)}$$

$$\mathcal{H} = \Pi \dot{\varphi} - \mathcal{L}$$

Hamiltonian density $\quad H = \int d^3x \, \mathcal{H}$

In our scalar Lagrangian density, we have

$$\Pi(x) = \dot{\varphi}(x)$$

$$\Rightarrow \mathcal{H} = \frac{1}{2}\Pi^2(x) + \frac{1}{2}(\vec{\nabla}\varphi(x))^2 + \frac{1}{2}m^2\varphi^2(x) - \Omega.$$

Don't this look familiar?

$$H = \frac{p^2}{2m} + \frac{1}{2}kx^2 \quad \text{(in QM)} \quad a_+, a_- \ldots$$

[1] Why: $\frac{d}{dt}$ of p. 7's integrand is $ie^{-ikx}(\partial_0^2 + \omega^2)\varphi$, and under $\int d^3x$, $\omega^2 = \mathbf{k}^2 + m^2 \to -\nabla^2 + m^2$ by parts. So $\dot{a} = i\int d^3x\, e^{-ikx}(-\partial^2 + m^2)\varphi = 0$ by KG. With interactions it is not zero and $a^\dagger(\mathbf{k})$ evolves; LecNote4 pp. 3–4 build LSZ on exactly this integral.

now we can get $\mathcal{H}$ in terms of $a(\vec{k})$, $a^*(\vec{k})$

$$+a^*(\vec{k})e^{-ikx}]$$

$$\mathcal{H} = \int d^3x\, \mathcal{H}' \qquad \leftarrow \int d^3x = V$$

$$= -\Omega_0 V + \frac{1}{2}\int \widetilde{dk}\,\widetilde{dk'}\,d^3x \Big\{$$

$$(-i\omega a(\vec{k})e^{-ikx} + i\omega a^*(\vec{k})e^{-ikx})(-i\omega' a(\vec{k'})e^{ik'x} + i\omega' a^*(\vec{k'})e^{-ik'x})$$

$$+ (+i\vec{k}a(\vec{k})e^{-ikx} - i\vec{k}a^*(\vec{k})e^{-ikx})(+i\vec{k'}a(\vec{k'})e^{ik'x} - i\vec{k'}a^*(\vec{k'})e^{-ik'x})$$

$$+ m^2(a(\vec{k})e^{-ikx} + a^*(\vec{k})e^{-ikx})(a(\vec{k'})e^{ik'x} + a^*(\vec{k'})e^{-ik'x})\Big\}$$

$$= -\Omega_0 V + \frac{1}{2}(2\pi)^3 \int \widetilde{dk}\,\widetilde{dk'}\Big\{$$

$$\delta(\vec{k}+\vec{k'})(-\omega\omega' - \vec{k}\cdot\vec{k'} + m^2)(a(\vec{k})a(\vec{k'})e^{-i(\omega+\omega')t} + a^*(\vec{k})a^*(\vec{k'})e^{i(\omega+\omega')t})$$

$$+\delta(\vec{k}-\vec{k'})(\omega\omega' + \vec{k}\cdot\vec{k'} + m^2)(a(\vec{k})a^*(\vec{k'})e^{-i(\omega-\omega')t}$$

$$+ a^*(\vec{k})a(\vec{k'})e^{i(\omega-\omega')t})\Big\}$$

$$= -\Omega_0 V + \frac{1}{2}\cdot\frac{1}{2\omega}\int \widetilde{dk}\Big\{$$

$$(-\omega^2 + \vec{k}^2 + m^2)(a(\vec{k})a(-\vec{k})e^{-2i\omega t} + a^*(\vec{k})a^*(-\vec{k})e^{2i\omega t})$$

$$(\omega^2 + \vec{k}^2 + m^2)(a(\vec{k})a^*(\vec{k}) + a^*(\vec{k})a(\vec{k}))\Big\}$$

$$= -\Omega_0 V + \frac{1}{2}\int \widetilde{dk}\,\omega\,(a(\vec{k})a^*(\vec{k}) + a^*(\vec{k})a(\vec{k}))$$

exactly what we want!

Relativistic QM (that reduce back to SE) with correct energy

And the theory is L.I. (from Lagrangian)

$$\int dx\, e^{-iax} = (2\pi)\,\delta(a)$$

$$\int d^3x\, e^{-i\vec{p}\cdot\vec{x}} = (2\pi)^3\,\delta^3(\vec{p})$$

The above are all classical field theory. Below, we start 2nd quantization.

We can take up the quantum theory now (from the classical equation of motion) through

Canonical Quantization.

(promoting q_i and p_i to operators, with

$$[q_i, q_j] = 0, \quad [p_i, p_j] = 0, \quad [q_i, p_j] = i\hbar\,\delta_{ij}$$

In Heisenberg Picture all operators are taken at Equal times.

$$[\varphi(\vec{x},t), \varphi(\vec{x}',t)] = 0$$

$$[\Pi(\vec{x},t), \Pi(\vec{x}',t)] = 0$$

$$[\varphi(\vec{x},t), \Pi(\vec{x}',t)] = i\hbar\,\delta^3(\vec{x}-\vec{x}')$$

~~need to~~ equivalently

Then we can have

$$[a(\vec{k}), a(\vec{k}')] = 0$$

$$[a^\dagger(\vec{k}), a^\dagger(\vec{k}')] = 0$$

$$[a(\vec{k}), a^\dagger(\vec{k}')] = (2\pi)^3\, 2\omega\, \delta^3(\vec{k}-\vec{k}') \quad (1)$$

1 Insert p. 7's $a(\mathbf{k}) = \int d^3x\, e^{-ikx}[\omega\varphi + i\Pi]$ and its adjoint ($\varphi = \varphi^\dagger$ now, so $a^* \to a^\dagger$): $[a(\mathbf{k}), a^\dagger(\mathbf{k}')] = \int d^3x\, e^{-i(k-k')x}(\omega + \omega') = (2\pi)^3 2\omega\,\delta^3(\mathbf{k}-\mathbf{k}')$; $[a,a]$ gets $\omega' - \omega = 0$ at $\mathbf{k}' = -\mathbf{k}$. The steps reverse, giving back $[\varphi, \Pi] = i\delta^3$: hence "equivalently".

Here $a^\dagger(\vec{k})$ replaced $a^*(\vec{k})$ for more general states.

Comment on $\Sigma^\circ V$ part.

Now

$$H = \int d\hat{k}\ \omega\, a^\dagger(\vec{k}) a(\vec{k}) + (\mathcal{E}_0 - \Omega_0) V$$

$$[a, a^\dagger] = i\delta^3(\vec{k} - \vec{k}')\,(2\pi)^3 \cdot 2\omega \,.$$ (1)

$$\mathcal{E}_0 V = \frac{1}{2} \int d\hat{k}\ \omega\, \delta^3(\vec{k} - \vec{k})\,(2\pi)^3 2\omega$$

infinite zero point energy.

1 No i: $[a(\mathbf{k}), a^\dagger(\mathbf{k}')] = (2\pi)^3 2\omega\, \delta^3(\mathbf{k} - \mathbf{k}')$, as on p. 10; $\langle 0|aa^\dagger|0\rangle$ is a norm, so it is real and positive. This page (hw 13) is a rougher pass over p. 12 (hw 12).

$$\mathcal{E}_0 = \frac{1}{2} \frac{1}{(2\pi)^3} \int d^3k\, \omega$$

as $\delta^3(\vec{0})(2\pi)^3 = \int d^3x\, e^{i\vec{0}\cdot\vec{x}} = \int d^3x = V.$

$e^{i\vec{0}\cdot\vec{x}}$

~~$\int d^3\hat{k}\, dk$~~

$$\int d^3k = \int_0^\Lambda d|k|\, d\Omega.$$ (2)

2 Should read $d^3k = |\mathbf{k}|^2\, d|\mathbf{k}|\, d\Omega$, as in p. 12's red side calculation.

if we integrate up to Λ.

which means we only work within a scale Λ.

$$\mathcal{E}_0 = \frac{1}{2} \times \frac{1}{2\pi^2} \cdot \Lambda^4.$$ (3)

3 Should read $E_0 = \frac{1}{2} \cdot \frac{1}{2\pi^2} \cdot \frac{\Lambda^4}{4} = \frac{\Lambda^4}{16\pi^2}$, as on p. 12: $\int_0^\Lambda k^3 dk = \Lambda^4/4$ with $\omega \simeq |\mathbf{k}|$. As written, E_0 is four times too large.

(12)

Now, a first encounter of ultraviolate divergence.

$$H = \int d\tilde{k}\, \omega\, a^\dagger(\vec{k})\, a(\vec{k}) + (\mathcal{E}_0 - \Omega_0) V \quad (1)$$

$$\mathcal{E}_0 = \frac{1}{2} \times \frac{1}{(2\pi)^3} \boxed{\int d^3k\, \omega} \simeq \int_0^\Lambda d|k|\, |\vec{k}|^2 \cdot d\Omega \cdot |\vec{k}| = \frac{4\pi}{4} |\vec{k}|^4 \Big|_0^\Lambda = \pi \Lambda^4.$$

we have interpreted $(2\pi)^3 \delta^{(3)}(\vec{0})$ as the volume of V

Recall $\int d^3x\, e^{-i\vec{q}\cdot\vec{x}} = (2\pi)^3 \delta^{(3)}(\vec{q})$.

$$\int d^3x\, e^{-i0} = \int d^3x = V$$

then $\mathcal{E}_0$ if we view there is a maximum of momentum we can integrate over to Λ

then $\mathcal{E}_0 = \dfrac{\Lambda^4}{16\pi^2}$

Λ is call the ultraviolate cutoff.

we choose $\Omega_0 = \mathcal{E}_0$ so the ground state has (2) zero energy.

About fermions, we should invoke anticommutation for both the field part and "spinor part".

1 Where E_0 comes from: as operators, $\frac{1}{2}\omega(aa^\dagger + a^\dagger a) = \omega a^\dagger a + \frac{1}{2}\omega[a, a^\dagger]$, and $[a(\mathbf{k}), a^\dagger(\mathbf{k})] = (2\pi)^3 2\omega\, \delta^3(\mathbf{0}) = 2\omega V$. So $E_0 V = \frac{1}{2}\int \widetilde{dk}\, \omega\, 2\omega V$: $\frac{1}{2}\omega$ per oscillator, $V\, d^3k/(2\pi)^3$ oscillators. $\Lambda^4/16\pi^2$ takes $\omega \simeq |\mathbf{k}|$ ($\Lambda \gg m$); the mass adds $m^2\Lambda^2/16\pi^2$.

2 This is p. 2's guess: with $a = (2\pi)^{3/2}\sqrt{2\omega}\,\tilde{a}$, $[\tilde{a}(\mathbf{k}), \tilde{a}^\dagger(\mathbf{k}')] = \delta^3(\mathbf{k} - \mathbf{k}')$ and $H = \int d^3k\, \omega\, \tilde{a}^\dagger \tilde{a}$ at $\Omega_0 = E_0$. Its invariance is now explicit: $U(\Lambda) a^\dagger(\mathbf{k}) U(\Lambda)^{-1} = a^\dagger(\Lambda k)$ is unitary because $\widetilde{dk}$ and $2\omega\, \delta^3(\mathbf{k} - \mathbf{k}')$ are invariant, and gives $U^{-1}\varphi(x)U = \varphi(\Lambda^{-1}x)$ (LecNote2 p. 12).